



Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Which sector commonly uses smart cards for identity verification?
A) Construction B) Education
C) Government and healthcare D) Entertainment
2. Smart cards can reduce fraud in which of the following systems?
A) Gaming systems B) Payment and banking systems
C) Television remotes D) Car engines
3. Which of the following is a type of smart card?
A) SIM card B) Debit card with magnetic stripe
C) Scratch card D) Greeting card
4. Which of the following communication methods is used by contactless smart cards?
A) Wi-Fi B) NFC (Near Field Communication)
C) Bluetooth D) Ethernet
5. One major benefit of Java Card Technology is the ability to?
A) Allow multiple applications to coexist securely B) Run only one fixed application per card
C) Eliminate cryptographic operations D) Avoid application updates
6. The Java Card Virtual Machine (JCVM) provides?
A) Direct hardware access without restrictions B) A runtime environment for executing Java Card bytecode
C) Support for floating-point operations D) Dynamic class loading during runtime
7. What is the primary function of a Card Acceptance Device (CAD)?
A) To generate encryption keys inside the smart card B) To manufacture and personalize smart cards
C) To store data permanently on the smart card D) To act as an interface between the smart card and host system
8. In a smart card communication model, which entity sends Application Protocol Data Units (APDUs)?
A) The smart card only B) The CAD only
C) The host computer (Application layer) D) The ISO/IEC standard body
9. A transient object in Java Card will lose its contents?
A) After power-off or card reset B) Only after explicit deletion
C) When the EEPROM gets overwritten D) When garbage collection runs
10. In Java Card, transient objects can be created in which memory?
A) EEPROM only B) RAM only
C) ROM only D) Flash only
11. Which class in Java Card provides methods for transaction management?
A) CardRuntime B) APDU
C) JCSystem D) Buffer

12. If an update to a persistent array must be crash-safe, which method should be used?
 A) JCSysm.arrayCopyNonAtomic() B) JCSysm.arrayFillNonAtomic()
 C) JCSysm D) JCSysm.arrayCopy()
13. Payment card type approval testing is primarily based on which standard?
 A) ISO/IEC 14443 B) ISO/IEC 7816 and EMV specifications
 C) PCI DSS D) IEEE 802.11
14. Which status word represents successful execution in ISO7816?
 A) 0x9000 B) 0x6D00
 C) 0x1231 D) 0x6F32
15. In the response APDU (as per ISO 7816), what do SW1 and SW2 represent?
 A) Command header values B) Response status words
 C) Status words indicating execution result D) Command parameters
16. What is the main purpose of the personalization phase of payment cards?
 A) Testing the durability of plastic B) Assigning PCI certification to the card issuer
 C) Printing holograms and logos D) Encoding cardholder data, keys, and applications onto the card
17. What is the main purpose of Global Platform Card Architecture?
 A) To provide a secure framework for application management on smart cards B) To certify POS terminals
 C) To manufacture chip hardware D) To define cardholder banking procedures
18. What is the purpose of logical channels in Global Platform cards?
 A) To increase APDU buffer size B) To allow multiple applications to be selected and communicated with simultaneously
 C) To decrease APDU buffer size D) To encrypt card communication
19. What is the main purpose of key management in Global Platform?
 A) To store cardholder PINs B) To increase card memory space
 C) To configure terminal hardware D) To manage secure channels, authentication, and application loading securely
20. Which command is used to remove applications, packages, or security domains from a card?
 A) DELETE B) LOCK
 C) INSTALL D) LOAD

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the various types of smart cards and their usage with necessary example.	08	L*2	1	1
	b) Define java card. Also, explain with a neat diagram the components of Java Card Virtual Machine (JCVM) architecture.	08	L2	2	1
2.	a) Define Smart Card. Also, explain the benefits and applications of a smart card.	08	L2	1	1
	b) Demonstrate with a neat diagram the components of java card overall architecture.	08	L2	2	1
3.	a) Explain the essential characteristics of smart card communication model.	08	L2	1	1
	b) Demonstrate the significance of various java card applet file formats supported by java card deployment.	08	L2	2	1
Unit – II					
4.	a) Define transient object. Also, explain the necessity of APDU class in the java card APIs.	08	L1, L2	3	1
	b) Explain the approval process of the payment cards with a neat diagram.	08	L2	4	1

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| 5. a) | Explain the transaction mechanism supported by java card technology. | 08 | L2 | 3 | 1 |
| b) | Illustrate EMV card / Chip architecture. | 08 | L2 | 4 | 1 |
| 6. a) | Summarize the essential properties of transient objects in java card technology. | 08 | L2 | 3 | 1 |
| b) | Explain the production and personalization of the payment cards with a neat diagram. | 08 | L2 | 4 | 1 |

Unit – III

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|-------|---|----|----|---|---|
| 7. a) | Explain along with INS about any four global platform (GP) commands. | 08 | L2 | 5 | 1 |
| b) | Illustrate with a neat diagram about card life cycle state transitions. | 08 | L2 | 5 | 1 |
| 8. a) | Illustrate along with INS about the necessity of content management and security management commands. | 08 | L2 | 5 | 1 |
| b) | Explain the process of secure card content management. | 08 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
